SBFL project report

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In this project, I try to work with JFreeChart data and find which methods look more suspicious. The data was a zip file that had 2,262 text files in it. Each file is one test case. The problem was that every single test was marked as passed so there was no failing tests at all. So what I did is I just manually picked 10 files and made them failed. I chose files 1, 5, 20, 50, 100, 200, 500, 750, 1000, and 1039. So in the end, I had 10 failed tests and 2,252 passed tests.

1- Running the program:

I wrote the program in Java. It reads all the txt files using BufferedReader. For each method I tracked two things, ef which is how many failed tests ran that method, and ep which is how many passed tests ran it.

After collecting the data, I calculate the suspiciousness scores using the four formulas **"Tarantula, SBI, Jaccard, and Ochiai"**

I created a class to store method name, ef, ep, and all scores. The results get saved automatically to sbfl_results.txt.

2- Results:

My program gave scores to 888 methods. When I looked at the top 10 for each formula I noticed the same methods kept showing up every time. Like **GanttRenderer:drawTask, GanttRenderer:drawItem,** and **CategoryLineAnnotation:setCategory2** were always at the top. All these methods had ef=1 and ep=0, which means they only showed up in failing tests and never in passing mode. Tarantula and SBI gave them a score of 1.0, Ochiai gave 0.316. The numbers are different, but all four formulas agree on which methods are the most suspicious which is a good sign.

3- Conclusion:

This project was interesting because we could actually see how SBFL works on a real program. I was able to read the coverage data, process it, and apply all four formulas to calculate scores for each method. Since all tests were originally passing, I had to manually choose some tests as failed, so the results are not from real bugs. Because of this, the output is more like a simulation and not actual fault detection. But still, the program works correctly and shows how the ranking is generated. It also shows that all formulas can produce similar top methods based on the data. Overall, I understand how SBFL works and how to use coverage data to generate suspiciousness scores.